

Research Summary

Hydroxymethylated Quinone Derivatives for Electrochemical CO₂ Capture

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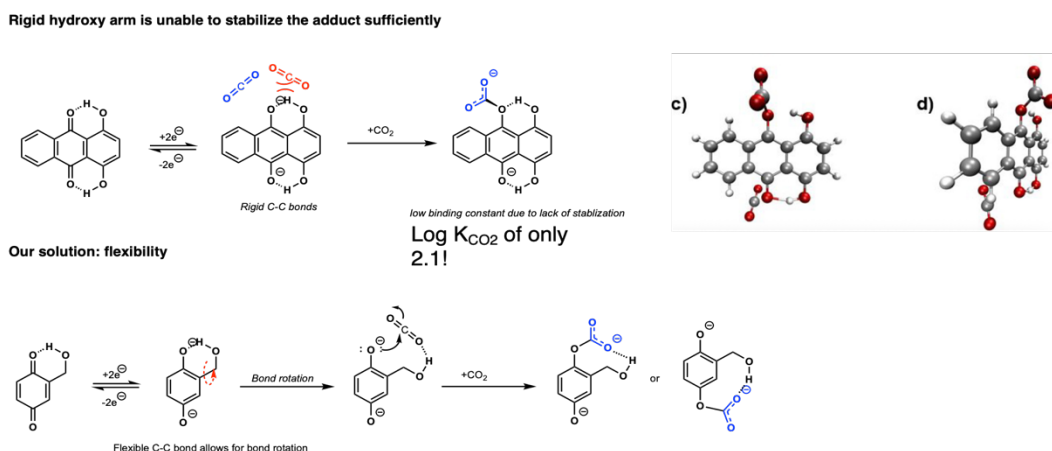
Background & Motivation

Electrochemical CO₂ capture using quinones is a promising alternative to energy-intensive amine scrubbing. Reduced quinone dianions bind CO₂ to form aryl-carbonate adducts, enabling reversible capture-release cycles driven by electricity. However, a fundamental trade-off limits these systems: the negative redox potentials needed to generate strongly nucleophilic dianions also increase oxygen sensitivity and side reactions. Prior work has shown that intermolecular hydrogen-bonding additives (e.g., alcohols) can stabilize reduced quinone intermediates and shift potentials anodically, but proximal H-bonding strategies where the donor is built into the quinone scaffold still remain largely unexplored for CO₂ capture.

Approach & Design Strategy

Prior work on intramolecular H-bonding in quinones for CO₂ capture used rigid α -hydroxy systems where the OH is directly bonded to the aromatic ring. In these rigid scaffolds, the OH group competes with CO₂ for the reduced carbonyl because it cannot reorient, resulting in poor CO₂ binding ($\log K_{\text{CO}_2} = 2.1$; Schimanofsky et al., J. Phys. Chem. C 2022). Our key design insight is that conformational flexibility solves this problem. We introduced a $-\text{CH}_2\text{OH}$ arm connected through a flexible benzylic C–C bond, which allows free bond rotation. We hypothesized in all oxidation state, the hydroxymethyl arm can provide favorable interaction (Scheme 1, bottom). Upon two-electron reduction, bond rotation can position the hydroxymethyl group to stabilize the dianion via intramolecular or intermolecular H-bonding. Critically, after CO₂ binding, the arm can further rotate to stabilize the resulting aryl-carbonate adduct, something rigid α -OH systems cannot achieve.

Scheme 1. Rigid vs. flexible H-bond donor design: rigid α -OH competes with CO₂ ($\log K = 2.1$), while the flexible $-\text{CH}_2\text{OH}$ arm is capable of rotating to stabilize the adduct.



I synthesized a series of hydroxymethylated quinones (HMQs A–D) based on this flexible design, along with chlorinated variants to further tune redox potentials via electron-withdrawing effects. Three compounds methoxymethyl derivative (MMQ, retains Oxygen atom but lacks H-bond donor) and a methyl derivative (MQ, lacks both) were selected to deconvolute hydrogen-bonding from inductive contributions.

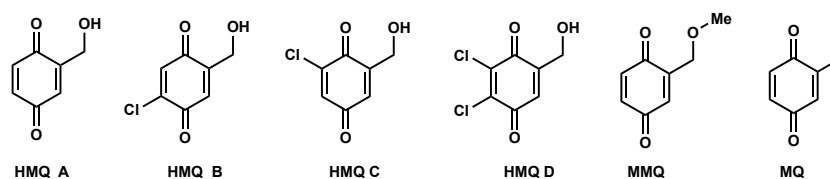
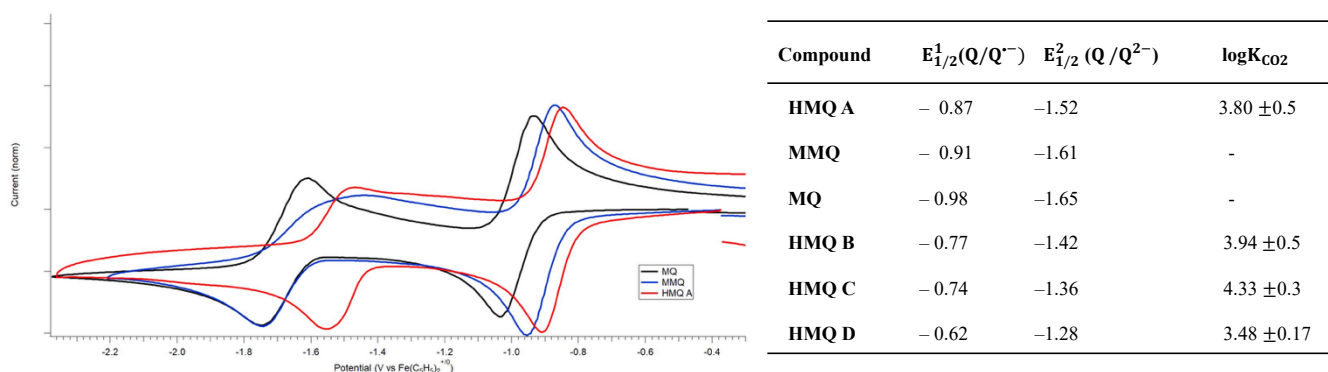


Chart 1. Molecular structures of HMQ A–D, MMQ, and MQ.

Key Findings

Redox Tuning via H-Bonding and Inductive Effects

Cyclic voltammetry in acetonitrile (0.2 M TBAPF₆) reveals that the hydroxymethyl group induces a 40–130 mV anodic shift in reduction potential (Figure 1 & Table 1). Comparison of HMQ A, MMQ, and MQ shows that hydrogen bonding and inductive effects operate through distinct mechanisms: the ether oxygen in MMQ preferentially stabilizes the semiquinone (Q^{•−}) via inductive withdrawal (70 mV shift in E_{1/2}¹ compared to MQ), while the hydroxyl in HMQ A predominantly stabilizes the dianion (Q^{2−}) through H-bonding (90 mV shift in E_{1/2}² compared to MMQ). Chlorine substitution provides an additional lever: monochlorinated HMQ B and C shift potentials 100–160 mV anodically, and the dichlorinated HMQ D achieves a 240 mV shift relative to the parent.



All potentials reported vs Fc^{+/0} in MeCN with 0.2 M TBAPF₆.

Figure 1. CVs of HMQ A (red), MMQ (blue), and MQ (black) in MeCN. Table 1. Summary of Redox Potentials

CO₂ Binding and Adduct Characterization

Spectroscopic and electrochemical evidence confirms clean formation of aryl-carbonate adducts upon CO₂ binding. The dianion of HMQ A was generated both chemically (TBAOMe deprotonation) and electrochemically (controlled potential electrolysis), with both routes yielding identical UV-vis signatures (λ_{max} 326, 430 nm for Q^{2−}; 303, 412 nm for the CO₂ adduct). ¹³C NMR at 158.7 ppm confirms carbonate formation. Crucially, CO₂ binding affinity is preserved across the chlorinated series despite large shifts in redox potential: log K(CO₂) = 3.80 (HMQ A), 3.94 (HMQ B), 4.49 (HMQ C), and 3.48 (HMQ D). This breaks the conventional linear trade-off between reduction potential and binding strength.

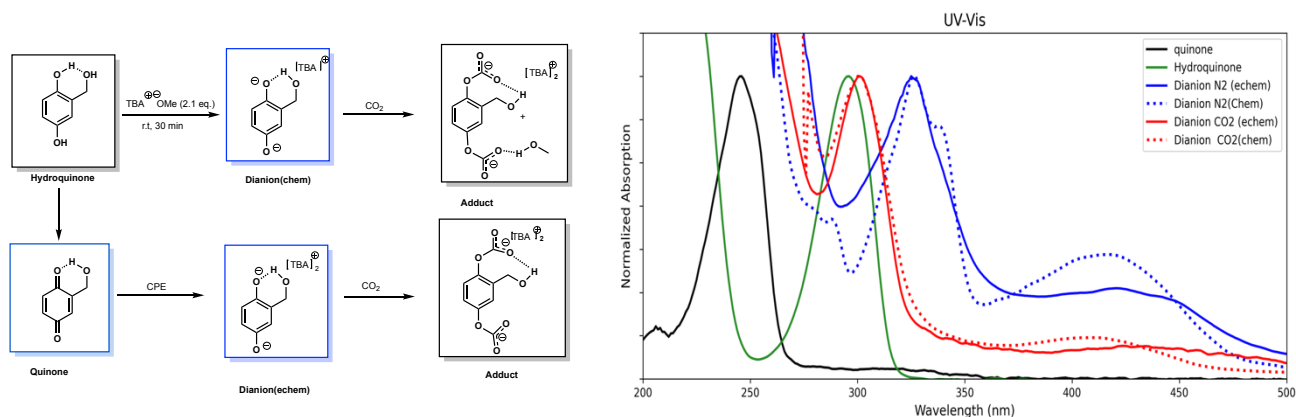


Figure 2. Uv-vis signatures of intermediate studied for this work

Significance & Outlook

This work demonstrates that proximal hydrogen bonding provides a complementary and generalizable design handle for tuning quinone redox properties without sacrificing CO₂-capture performance. The combination of proximal H-bond donors with electron-withdrawing substituents enables independent control over reduction potential and binding affinity—circumventing the fundamental linear trade-off that has constrained quinone-based capture systems.